Задание 1

1. $z = \frac{2}{1+i} - \frac{(1+i)(2-2i)}{(1-i)(1-2i)} = \frac{2}{1+i} - \frac{3}{-1-3i} = \frac{2}{1+i} + \frac{3}{1+3i} =$

$= \frac{2+6i+3+3i}{1+3i+i-3} = \frac{5+9i}{-2+4i} \cdot \frac{-2-4i}{-2-4i} = \frac{-10-20i-18i+36}{4+8i-8i+16} =$

$= \frac{26-38i}{20} = 1,3 - 1,9i$

2. $z = \frac{\sqrt{3}+i}{2-i\sqrt{5}} - \frac{(1+i)(\sqrt{3}+2i)}{7} = \frac{2\sqrt{3}+i\sqrt{15}+2i-\sqrt{5}}{4+(\sqrt{5})^2} - \frac{\sqrt{3}+2i+i\sqrt{3}-2}{7} =$

$= \frac{14\sqrt{3}+i\cdot 7\sqrt{15}+14i-\sqrt{5}\cdot 7-9\sqrt{3}-18i-i\cdot 9\sqrt{3}+18}{7\cdot 9} =$

$= \frac{5\sqrt{3}+18-7\sqrt{5}}{63} + i\,\frac{7\sqrt{15}-9\sqrt{3}-4}{63}$

3. $z = 1 + i\sqrt{3}$ $\quad Re\,z = x = 1$

$Im\,z = y = \sqrt{3}$

$|z| = \sqrt{x^2+y^2} = \sqrt{1+3} = \sqrt{4} = 2$; $\varphi = arctg\frac{y}{x} = arctg\sqrt{3} = \frac{\pi}{3}$

$z = |z|\cdot e^{i(\varphi+2\pi n)} = 2\cdot e^{\frac{\pi i}{3}+2\pi i n}$

$z = |z|\cdot(\cos(\varphi+2\pi n) + i\sin(\varphi+2\pi n)) = 2(\cos(\frac{\pi}{3}+2\pi n) + i\sin(\frac{\pi}{3}+2\pi n))$

4. $z = -1 + i\sqrt{3}$ $\quad x = -1$; $y = \sqrt{3}$

$|z| = 2$; $\varphi = arctg(-\sqrt{3}) = -\frac{\pi}{3}$

$z = 2e^{i(-\frac{\pi}{3}+2\pi n)} = 2(\cos(-\frac{\pi}{3}+2\pi n) + i\sin(-\frac{\pi}{3}+2\pi n))$

5. $z = -1 - i\sqrt{3}$ $\quad x = -1$ $\quad y = -\sqrt{3}$

$|z| = 2$

$\varphi = arctg(\frac{-\sqrt{3}}{-1}) = \frac{\pi}{3}$

$z = 2e^{i(\frac{\pi}{3}+2\pi n)} = 2(\cos(\frac{\pi}{3}+2\pi n) + i\sin(\frac{\pi}{3}+2\pi n))$

6. $z=\sqrt{3}-i$ $\quad x=\sqrt{3}$

$y=-1$

$|z|=2$

$\varphi=\operatorname{arctg}\left(\frac{-1}{\sqrt{3}}\right)=-\frac{\pi}{6} \Rightarrow z=2e^{-i\left(\frac{\pi}{6}+2\pi n\right)}=2\left(\cos\left(-\frac{\pi}{6}+2\pi n\right)+i\sin\left(-\frac{\pi}{6}+2\pi n\right)\right)$

7. $z^4-zi=0$

$z(z^3-i)=0$

$z=0$ или $z=\sqrt[3]{i}$ $\quad x=0$

$y=1$

$\rho=\sqrt{x^2+y^2}=1$

$\varphi=\operatorname{arctg}\frac{1}{0}=\frac{\pi}{2}$

$z=\cos\left(\frac{\frac{\pi}{2}+2\pi n}{3}\right)+i\sin\left(\frac{\frac{\pi}{2}+2\pi n}{3}\right)$

8. $z^6-8=0$

$z^6=8 \Rightarrow z=\pm\sqrt[6]{8}$

9. $z^4+zi=0$

$z=0$ или $z^3+i=0$

$z=\sqrt[3]{-i}$ $\quad x=0;\ y=-1$

$\rho=1$

$\varphi=\operatorname{arctg}\left(\frac{-1}{0}\right)=\frac{\pi}{2}$

$z=\cos\left(\frac{\frac{\pi}{2}+2\pi n}{3}\right)+i\sin\left(\frac{\frac{\pi}{2}+2\pi n}{3}\right)$

10. $z^5+z=0$

$z(z^4+1)=0$

$z=0$ или $z=\sqrt[4]{-1}$ $\quad x=-1$

$y=0$

$\rho=1$

$\varphi=\operatorname{arctg}\left(\frac{0}{-1}\right)=0$

$z=\cos\left(\frac{2\pi n}{4}\right)+i\cdot\sin\left(\frac{2\pi n}{4}\right)=\cos\frac{\pi n}{2}+i\sin\frac{\pi n}{2}$

Задание 2

1. $f(z) = \frac{1}{z^2(z-3)} = \frac{A}{z} + \frac{B}{z^2} + \frac{C}{z-3} = \frac{Az^2 - 3Az + Bz - 3B + Cz^2}{z^2(z-3)}$

$0 < |z| < 3 \Rightarrow |z| < 3 \Rightarrow \frac{3}{|z|} > 1 \Rightarrow \frac{|z|}{3} < 1$

Приравниваем коэффициенты:

$$\begin{matrix} z^2: \\ z: \\ z^0: \end{matrix}\begin{cases} A + C = 0 \\ -3A + B = 0 \\ -3B = 1 \end{cases} \quad \begin{cases} C = \frac{1}{9} \\ A = \frac{B}{3} = -\frac{1}{9} \\ B = -\frac{1}{3} \end{cases}$$

$$f(z) = -\frac{1}{9}\cdot\frac{1}{z} - \frac{1}{3}\cdot\frac{1}{z^2} + \frac{1}{9}\cdot\frac{1}{z-3} = \frac{-1}{9z} - \frac{1}{3z^2} + \frac{1}{9}\cdot\frac{-1}{\left(1-\frac{z}{3}\right)3} =$$

$$= \frac{-1}{9z} - \frac{1}{3z^2} - \frac{1}{9\cdot 3}\cdot\sum_{n=0}^{\infty}\left(\frac{z}{3}\right)^n$$

2. $f(z) = \frac{1}{z^2(z-3)}$ в окр. $z = \infty$

$$f(z) = \frac{-1}{9z} - \frac{1}{3z^2} + \frac{1}{9}\cdot\frac{1}{z\left(1-\frac{3}{z}\right)} = \frac{-1}{9z} - \frac{1}{3z^2} + \frac{1}{9z}\cdot\sum_{n=0}^{\infty}\left(\frac{3}{z}\right)^n$$

3. $f(z) = \frac{z+1}{z^2(z-1)} = \frac{A}{z} + \frac{B}{z^2} + \frac{C}{z-1} = \frac{Az^2 - Az + Bz - B + Cz^2}{z^2(z-1)}$

$$\begin{matrix} z^2: \\ z: \\ z^0: \end{matrix}\begin{cases} A + C = 0 \\ -A + B = 1 \\ -B = 1 \end{cases} \quad \begin{cases} C = 2 \\ A = -2 \\ B = -1 \end{cases} \Rightarrow f(z) = \frac{-2}{z} - \frac{1}{z} + 2\cdot\frac{1}{z-1} =$$

$$= \frac{-2}{z} - \frac{1}{z^2} + 2\cdot\frac{-1}{1-z} = \frac{-2}{z} - \frac{1}{z^2} - 2\cdot\sum_{n=0}^{\infty} z^n$$

4. $$f(z) = \frac{-2}{z} + \frac{-1}{z^2} + 2\cdot\frac{1}{z\left(1-\frac{1}{z}\right)} = \frac{-2}{z} - \frac{1}{z^2} + \frac{2}{z}\sum_{n=0}^{\infty}\left(\frac{1}{z}\right)^n$$

6. $f(z) = \frac{4z}{(z-1)^2\cdot(z+2)} = \frac{A}{z+2} + \frac{B}{z-1} + \frac{C}{(z-1)^2} =$

$$= \frac{Az^2 - 2Az + A + B(z^2 + z - 2) + Cz + 2C}{(z+2)(z-1)^2}$$

z^2: $A+B=0$

z: $-2A+B+C=4$

z^0: $A-2B+2C=0$

$$\begin{cases} A=-B \\ 3B+C=4 \\ -3B+2C=0 \end{cases}$$

$3C=4 \Rightarrow C=\frac{4}{3}$; $B=\frac{4-C}{3}=\frac{4-\frac{4}{3}}{3}=\frac{8}{9}$

$A=-\frac{8}{9}$

$$f(z)=-\frac{8}{9}\cdot\frac{1}{z+2}+\frac{8}{9}\cdot\frac{1}{z-1}+\frac{4}{3}\left(\frac{1}{z-1}\right)^2$$

Кольцо: $0<|z-1|<3$

$-3<z-1<3$

$2<z<4 \Rightarrow \frac{1}{z+2}=\frac{1}{\left(1+\frac{2}{z}\right)z}$

$\frac{1}{z-1}=\frac{1}{z\left(1-\frac{1}{z}\right)}$

$$f(z)=\frac{-8}{9}\cdot\sum_{n=0}^{\infty}\left(\frac{-2}{z}\right)^n+\frac{8}{9}\cdot\sum_{n=0}^{\infty}\frac{1}{z}\cdot\left(\frac{1}{z}\right)^n+\frac{4}{3}\sum_{n=0}^{\infty}\frac{2(2-1)\cdot\left(\frac{1}{z}\right)^n}{n!}$$

6. В окр. $z=\infty$

$$f(z)=\frac{-8}{9}\sum_{n=0}^{\infty}(-1)^n\cdot\left(\frac{2}{z}\right)^n+\frac{8}{9}\sum_{n=0}^{\infty}\frac{1}{z}\cdot\left(\frac{1}{z}\right)^n+\frac{4}{3}\sum_{n=0}^{\infty}\frac{2}{n!}\cdot\frac{1}{z^n}$$

7. $$f(z)=\frac{2z}{(z-1)(z+2)}=\frac{A}{z-1}+\frac{B}{z+2}=\frac{Az+2A+Bz-B}{(z-1)(z+2)}$$

$$\begin{cases} A+B=2 \\ 2A-B=0 \end{cases} \quad \begin{cases} 3A=2 \\ B=2A \end{cases} \quad \begin{cases} A=\frac{2}{3} \\ B=\frac{4}{3} \end{cases}$$

$$f(z)=\frac{2}{3}\cdot\frac{1}{z-1}+\frac{4}{3}\cdot\frac{1}{z+2}=\frac{2}{3}\cdot\frac{1}{z\left(1-\frac{1}{z}\right)}+\frac{4}{3}\cdot\frac{1}{2\left(1+\frac{z}{2}\right)}=$$

$$=\frac{2}{3}\sum_{n=0}^{\infty}\frac{1}{z}\cdot\left(\frac{1}{z}\right)^n+\frac{2}{3}\sum_{n=0}^{\infty}\left(\frac{-z}{2}\right)^n$$

8. $$f(z)=\frac{2}{3}\cdot\frac{1}{z\left(1-\frac{1}{z}\right)}+\frac{4}{3}\frac{1}{z\left(1+\frac{2}{z}\right)}=\frac{2}{3}\sum_{n=0}^{\infty}\frac{1}{z^{n+1}}+\frac{4}{3}\sum_{n=0}^{\infty}\frac{1}{z}\left(\frac{-2}{z}\right)^n$$

9. $\underset{z=\infty}{f(z)} = \dfrac{1}{(z^2+1)^2} + \sin\dfrac{1}{z} + 6z^5 = \sum\limits_{n=0}^{\infty} \dfrac{2(2-1)\dots(2-n+1)}{n!} \cdot z^{2n} +$

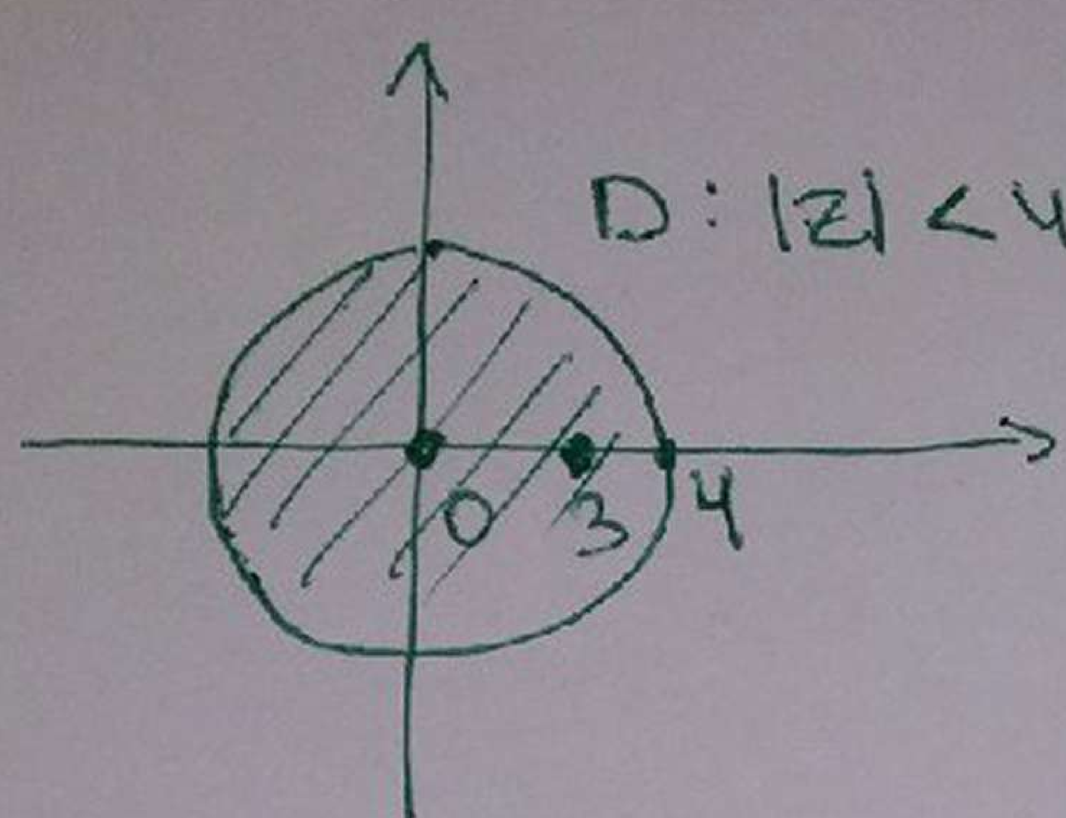
$+ \sum\limits_{n=1}^{\infty} \dfrac{(-1)^{n+1}}{(2n-1)!} \cdot \dfrac{1}{z^n} + 6z^5$

10. $\underset{z=0}{f(z)} = \dfrac{1}{(z^2+1)^2} + \sin\dfrac{1}{z} + 6z^5 = \sum\limits_{n=0}^{\infty} \dfrac{2(2-1)\dots(2-n+1)}{n!} \cdot z^{2n} +$

$+ \sum\limits_{n=1}^{\infty} \dfrac{(-1)^{n+1}}{(2n-1)!} \cdot \left(\dfrac{1}{z}\right)^n + 6z^5$

Задание 3

3.1 $\displaystyle\int\limits_{\partial D} \frac{1}{z^2(z-3)}\,dz \;=$



$0 \in D$

$3 \in D$

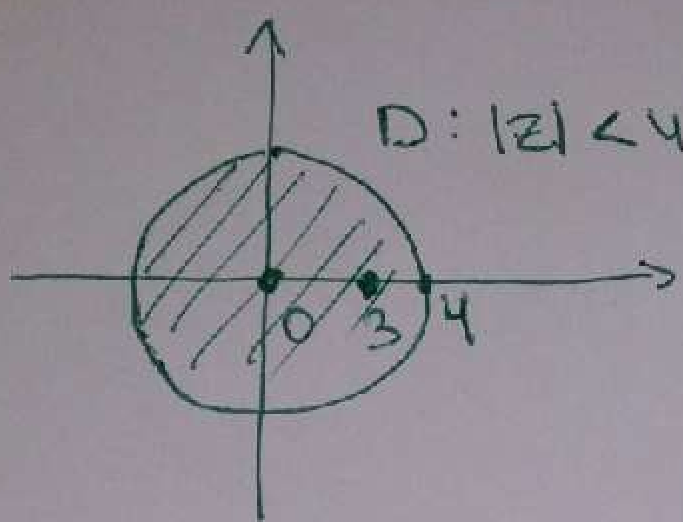
$f(z) = \dfrac{1}{z^2(z-3)}$

$\underset{z=\infty}{\operatorname{res}} f(z) = 0$, т.к. $f(\infty) = 0$

$= 2\pi i \cdot \left(\underset{z=0}{\operatorname{res}} f(z) + \underset{z=3}{\operatorname{res}} f(z) \right) =$

$= 2\pi i \left(-\underset{z=\infty}{\operatorname{res}} f(z) \right) = 2\pi i \cdot 0 = 0$

3.1 $\displaystyle\int_{\partial D} \frac{1}{z^2(z-3)}\,dz \ominus$



$0 \in D$

$3 \in D$

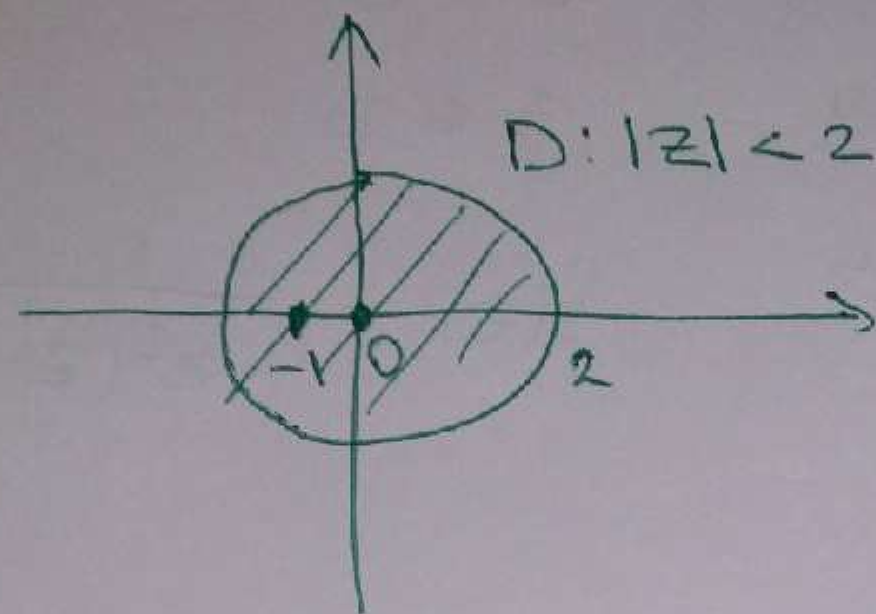
$$f(z) = \frac{1}{z^2(z-3)}$$

$$\operatorname*{res}_{z=\infty} f(z) = 0, \quad \text{т.к. } f(\infty) = 0$$

$$\ominus\ 2\pi i \cdot \left(\operatorname*{res}_{z=0} f(z) + \operatorname*{res}_{z=3} f(z)\right) =$$

$$= 2\pi i \left(-\operatorname*{res}_{z=\infty} f(z)\right) = 2\pi i \cdot 0 = 0$$

3.2 $\displaystyle\oint_{\partial D} \frac{\sin\frac{2}{z}}{1+z}\,dz \ominus$



$-1 \in D$

$0 \in D$

$$f(z) = \frac{\sin\frac{2}{z}}{1+z}$$

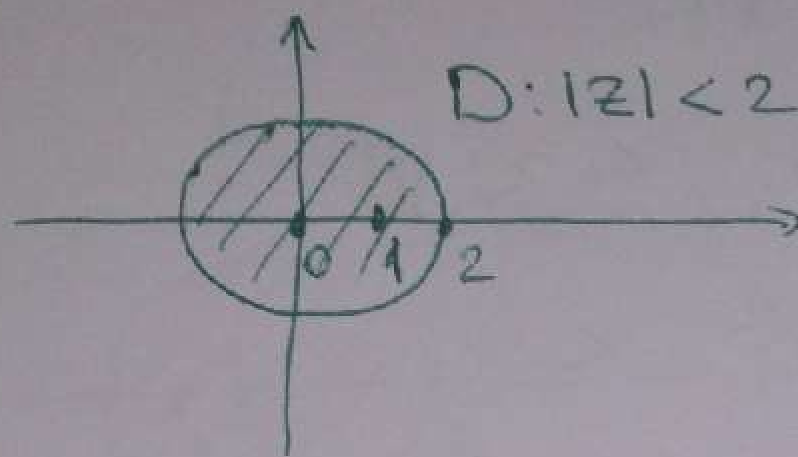
$$\operatorname*{res}_{z=\infty} f(z) = 0, \quad \text{т.к. } f(\infty) = 0$$

$$\ominus\ 2\pi i \cdot \left(\operatorname*{res}_{z=-1} f(z) + \operatorname*{res}_{z=0} f(z)\right) =$$

$$= 2\pi i \left(-\operatorname*{res}_{z=\infty} f(z)\right) = 2\pi i \cdot (0) = 0$$

3.3

$$\int_{\partial D} \frac{z+1}{z^2(z-1)}\,dz \ominus$$



$0 \in D$

$1 \in D$

$$f(z) = \frac{z+1}{z^2(z-1)}$$

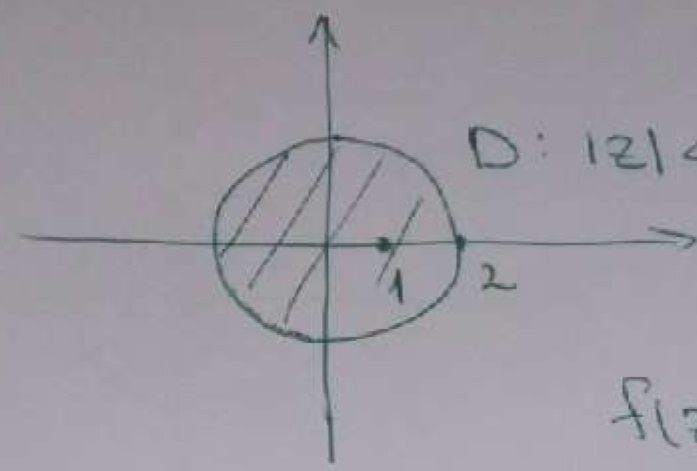
$$f(\infty) = 0$$

$$\operatorname*{res}_{z=\infty} f(z) = 0$$

$$\ominus\ 2\pi i \cdot \left(\operatorname*{res}_{z=0} f(z) + \operatorname*{res}_{z=1} f(z) \right) =$$

$$= 2\pi i \cdot \left(-\operatorname*{res}_{z=\infty} f(z) \right) = 2\pi i \cdot 0 = 0$$

3.4 $\displaystyle\int_{\partial D} z \cdot \sin \frac{z+1}{z-1}\, dz$



$D: |z| < 2$

$1 \in D$

$f(z) = z \cdot \sin \frac{z+1}{z-1}$

$$\lim_{z \to \infty} z \cdot \sin \frac{z+1}{z-1} = \infty$$

$$f(z) = z \cdot \sin\left(1 + \frac{2}{z-1}\right) =$$

$$= z \cdot \left(\sin 1 \cdot \cos \frac{2}{z-1} + \cos 1 \cdot \sin \frac{2}{z-1}\right) =$$

$$= z \cdot \left(\sin 1 \cdot \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} \cdot \left(\frac{2}{z-1}\right)^{2n} + \cos 1 \cdot \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} \cdot \left(\frac{2}{z-1}\right)^{2n-1}\right) =$$

$$= ((z-1)+1) \cdot \left(\sin 1 \cdot \sum_{n=0}^{\infty} \frac{(-1)^n \cdot 4^n}{(2n)!} \cdot \frac{1}{(z-1)^{2n}} + \cos 1 \cdot \sum_{n=0}^{\infty} \frac{(-1)^n \cdot 2^{2n-1}}{(2n+1)!} \cdot \frac{1}{(z-1)^{2n-1}} \right) =$$

$$= \sin 1 \cdot \left(\sum_{n=0}^{\infty} \frac{(-1)^n \cdot 4^n}{(2n)!} \cdot \frac{1}{(z-1)^{2n-1}} + \sum_{n=0}^{\infty} \frac{(-1)^n \cdot 4^n}{(2n)!} \cdot \frac{1}{(z-1)^{2n}} \right)$$

$$+ \cos 1 \cdot \left(\sum_{n=0}^{\infty} \frac{(-1)^n \cdot 2^{2n-1}}{(2n+1)!} \cdot \frac{1}{(z-1)^{2n-2}} + \sum_{n=0}^{\infty} \frac{(-1)^n \cdot 2^{2n-1}}{(2n+1)!} \cdot \frac{1}{(z-1)^{2n-1}} \right)$$

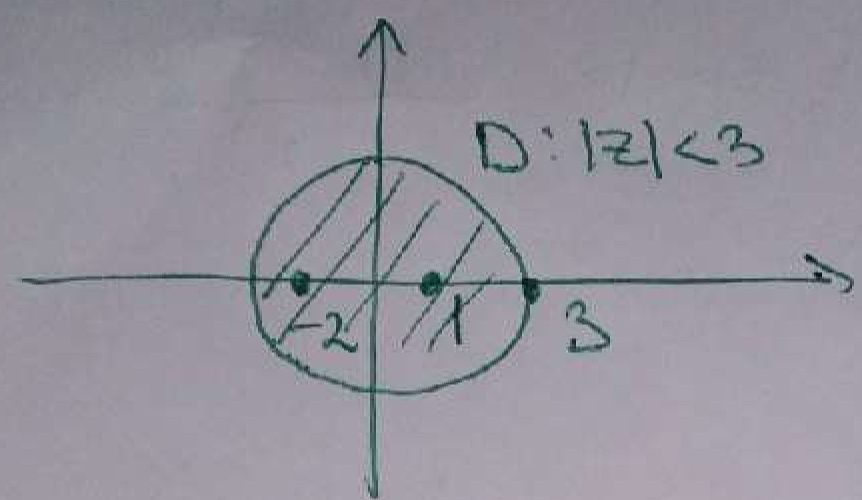
$$\Rightarrow \int_{\partial D} z \cdot \sin\frac{z+1}{z-1}\,dz = 2\pi i \cdot \operatorname*{res}_{z=1} f(z) =$$

$$= 2\pi i \cdot c_{-1} = 2\pi i \cdot \left(\sin 1 \cdot \frac{(-1)^1 \cdot 4^1}{2!} + \right.$$

$$\left. + \cos 1 \cdot \frac{(-1)^1 \cdot 2^1}{3!} \right) =$$

$$= 2\pi i \cdot \left(-2 \sin 1 - \frac{1}{3} \cos 1 \right)$$

3.5 $\displaystyle\int_{\partial D} \frac{4z}{(z-1)^2(z+2)}\,dz =$



$-2 \in D$

$1 \in D$

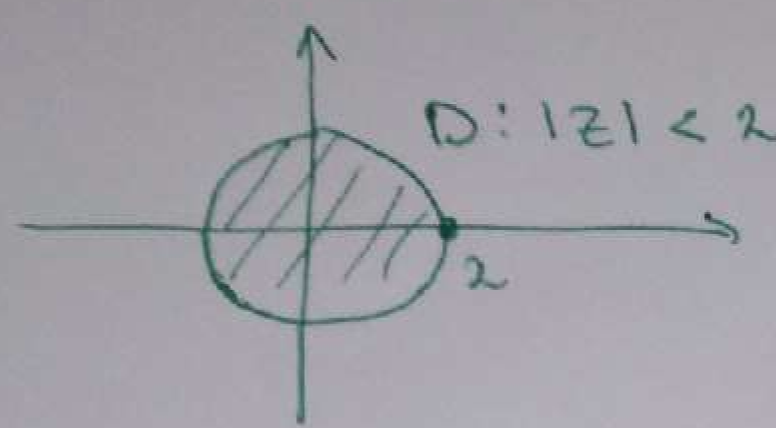
$$f(z) = \frac{4z}{(z-1)^2(z+2)}$$

$\operatorname*{res}_{z=\infty} f(z) = 0$, т.к. $f(\infty) = 0$

$$= 2\pi i \cdot \left(\operatorname*{res}_{z=-2} f(z) + \operatorname*{res}_{z=1} f(z) \right) =$$

$$= 2\pi i \cdot \left(-\operatorname*{res}_{z=\infty} f(z) \right) = 2\pi i \cdot 0 = 0$$

3.6 $\displaystyle\int_{\partial D} \frac{z^3}{z^4-1}\,dz \ominus$



$$f(z) = \frac{z^3}{z^4-1} =$$

$$= \frac{z^3}{(z^2-1)(z^2+1)}$$

$z = \pm 1 \in D$

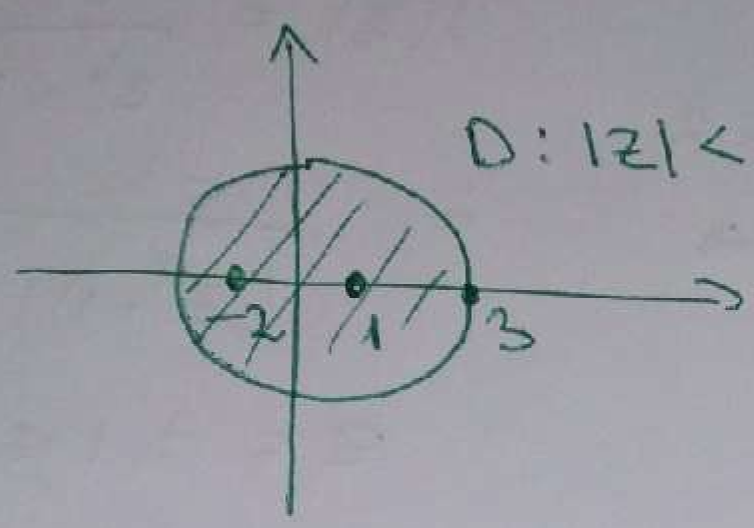
$z = \pm i \in D$

$$\operatorname*{res}_{z=\infty} f(z) = 0, \text{ т.к. } f(\infty) = 0$$

$$\ominus\ 2\pi i \cdot \left(\operatorname*{res}_{z=1} f(z) + \operatorname*{res}_{z=-1} f(z) + \operatorname*{res}_{z=i} f(z) + \operatorname*{res}_{z=-i} f(z)\right) =$$

$$= 2\pi i \cdot \left(-\operatorname*{res}_{z=\infty} f(z)\right) = 2\pi i \cdot 0 = 0$$

3.7 $\displaystyle\int_{\partial D} \frac{2z}{(z-1)(z+2)}\,dz \ominus$



$D : |z| < 3$

$-2 \in D$

$1 \in D$

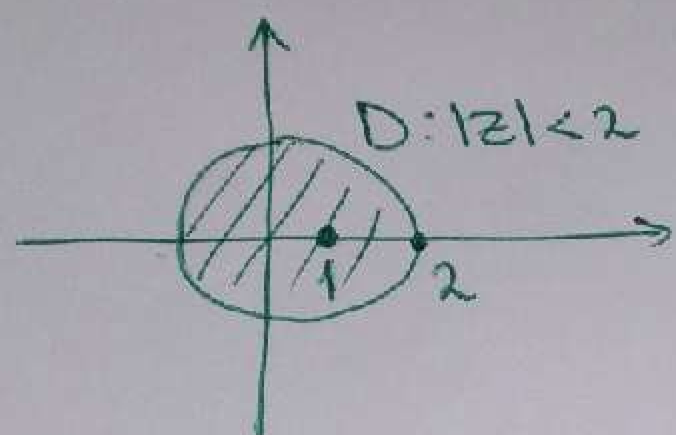
$f(z) = \dfrac{2z}{(z-1)(z+2)}$

$\operatorname*{res}_{z=\infty} f(z) = 0$, т.к. $f(\infty) = 0$

$\ominus\ 2\pi i \cdot (\operatorname*{res}_{z=-2} f(z) + \operatorname*{res}_{z=1} f(z)) =$

$= 2\pi i \cdot (-\operatorname*{res}_{z=\infty} f(z)) = 2\pi i \cdot 0 = 0$

3.8 $\displaystyle\int\limits_{\partial D} \frac{z}{(z-1)(z^2-2)}\,dz \; \circled{=}$



$z = 1 \in D$

$z = \pm\sqrt{2} \in D$

$$f(z) = \frac{z}{(z-1)(z^2-2)}$$

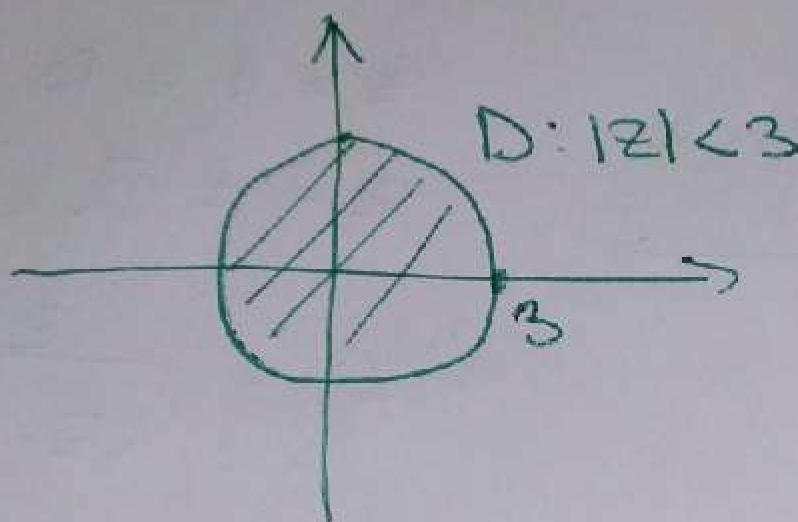
$\operatorname*{res}\limits_{z=\infty} f(z) = 0$, т.к. $f(\infty) = 0$

$$= 2\pi i \cdot \left(\operatorname*{res}_{z=1} f(z) + \operatorname*{res}_{z=\sqrt{2}} f(z) + \operatorname*{res}_{z=-\sqrt{2}} f(z) \right) =$$

$$= 2\pi i \cdot \left(-\operatorname*{res}_{z=\infty} f(z) \right) =$$

$$= 2\pi i \cdot 0 = 0$$

3.9 $\displaystyle\oint_{\partial D} \frac{z}{(z^2+2)(z-2)}\,dz \ominus$



$z = \pm\sqrt{2}i \in D$

$z = 2 \in D$

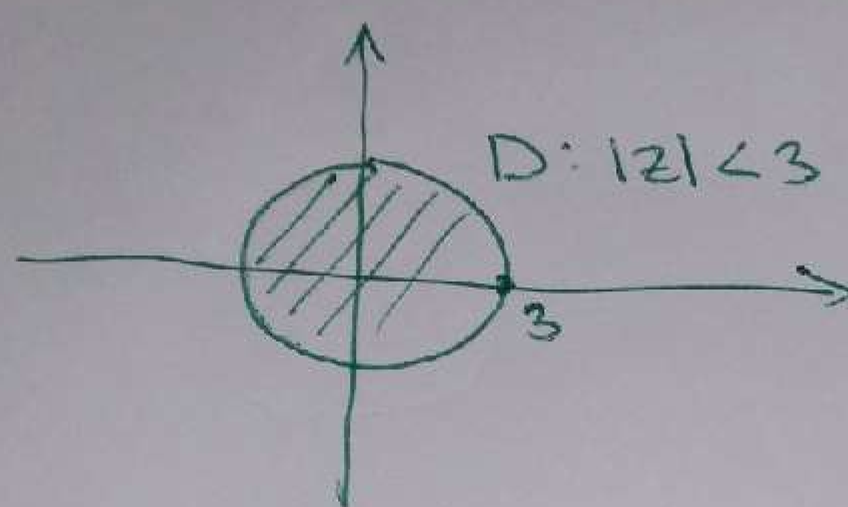
$$f(z) = \frac{z}{(z^2+2)(z-2)}$$

$$\operatorname*{res}_{z=\infty} f(z) = 0, \text{ т.к. } f(\infty) = 0$$

$$\ominus\ 2\pi i \cdot \left(\operatorname*{res}_{z=\sqrt{2}i} f(z) + \operatorname*{res}_{z=-\sqrt{2}i} f(z) + \operatorname*{res}_{z=2} f(z)\right) =$$

$$= 2\pi i \cdot \left(-\operatorname*{res}_{z=\infty} f(z)\right) = 2\pi i \cdot 0 = 0$$

3.10 $\displaystyle\int_{\partial D} \frac{z}{(z^2+1)(z+2)}\,dz =$



$z = \pm i \in D$

$z = -2 \in D$

$$f(z) = \frac{z}{(z^2+1)(z+2)}$$

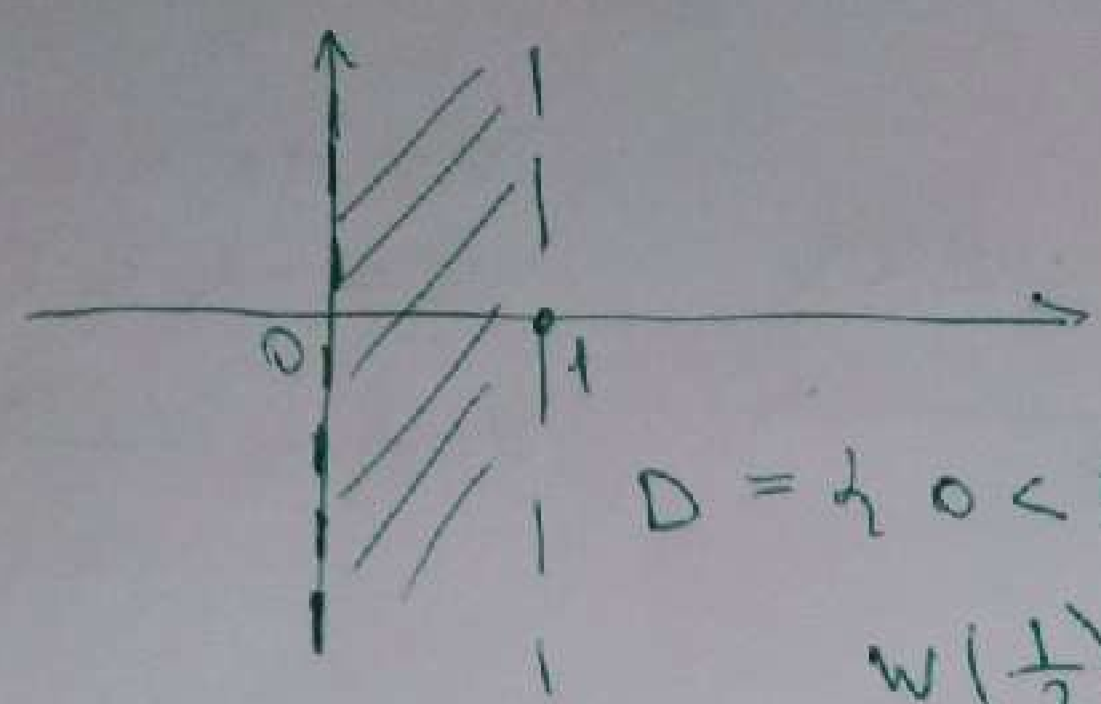
$\operatorname*{res}_{z=\infty} f(z) = 0$, т.к. $f(\infty) = 0$

$$= 2\pi i \cdot \left(\operatorname*{res}_{z=i} f(z) + \operatorname*{res}_{z=-i} f(z) + \operatorname*{res}_{z=-2} f(z)\right) =$$

$$= 2\pi i \cdot \left(-\operatorname*{res}_{z=\infty} f(z)\right) = 2\pi i \cdot 0 = 0$$

Задание 4

ч.1 $w = \frac{z-1}{z+2}$



$D = \{0 < Re\, z < 1\}$

$w\left(\frac{1}{2}\right) = \frac{-\frac{1}{2}}{\frac{5}{2}} = -\frac{1}{5}$

$w(0 + y\cdot i) = \frac{yi-1}{yi+2} = \frac{(yi-1)(yi-2)}{-y^2-4} =$

$= \frac{2-y^2+3yi}{-y^2-4} = \frac{y^2-2}{y^2+4} + \frac{3y}{y^2+4}i$

$w(0) = -\frac{1}{2}, \quad w(\infty) = 1$

$z_0 = \frac{-\frac{1}{2}+1}{2} = \frac{1}{4}$

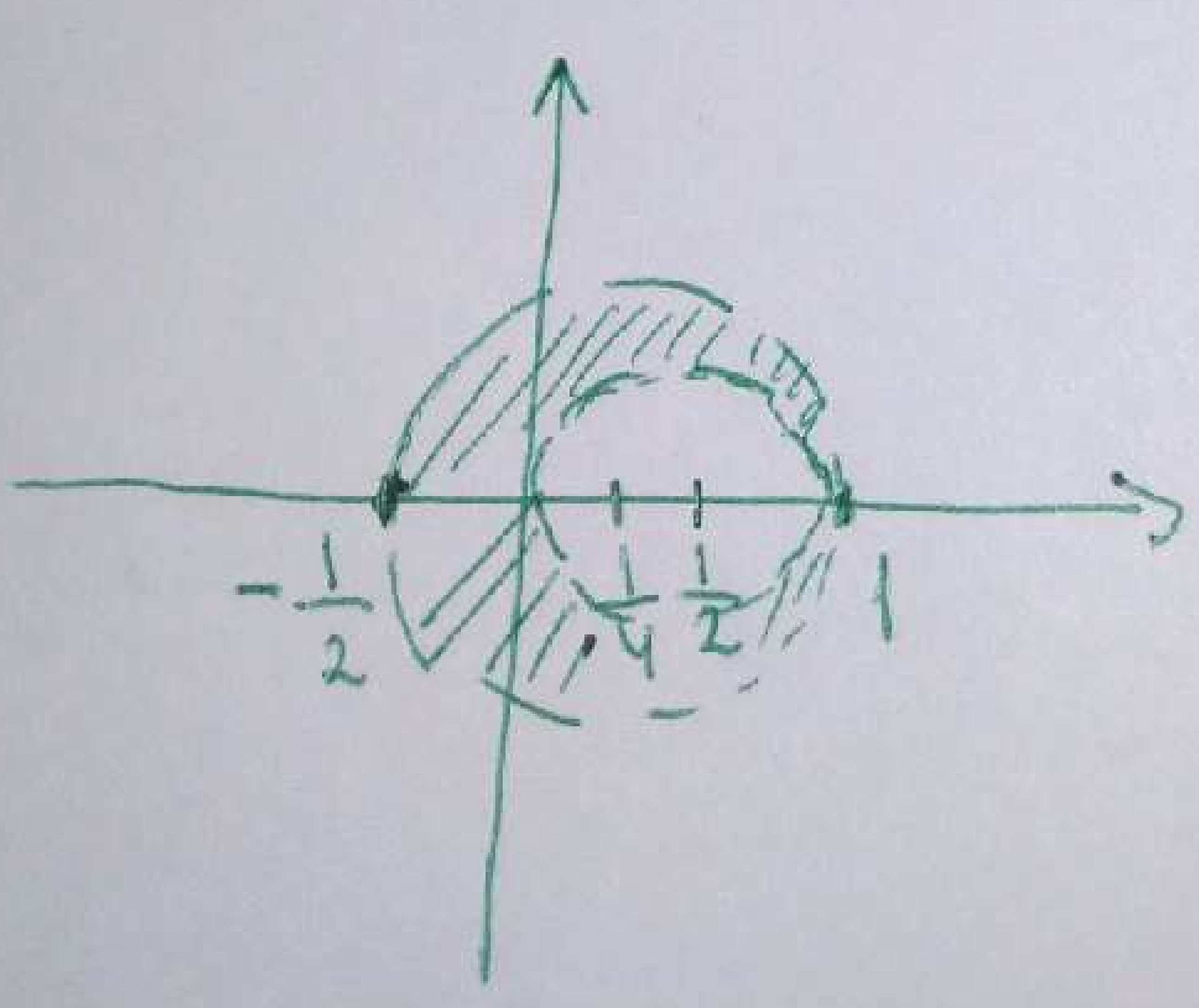
$r = \frac{1+\frac{1}{2}}{2} = \frac{3}{4}$

$\Rightarrow w: Re\, z = 0 \longrightarrow |z - \frac{1}{4}| = \frac{3}{4}$

$w(1) = 0, \quad w(\infty) = 1$

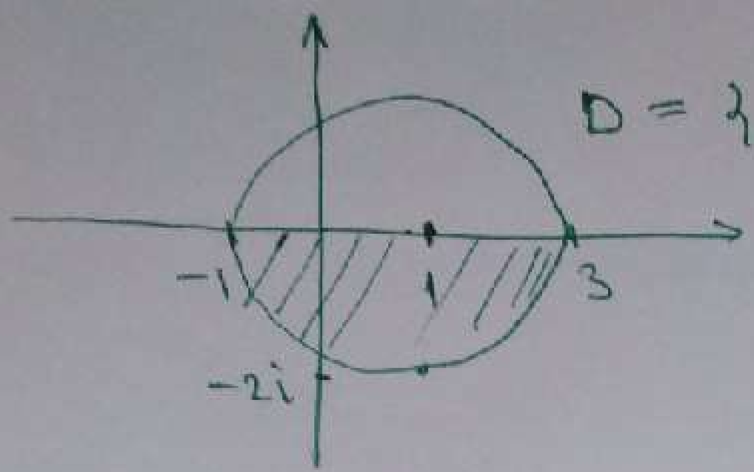
$z_0 = \frac{0+1}{2} = \frac{1}{2}, \quad r = \frac{1-0}{2} = \frac{1}{2}$

$\Rightarrow w: Re\, z = 1 \longrightarrow |z - \frac{1}{2}| = \frac{1}{2}$



$\{|z - \frac{1}{4}| < \frac{3}{4}, |z - \frac{1}{2}| > \frac{1}{2}\}$

4.2 $\quad w = \dfrac{z+1}{z-3}$



$D = \{ |z-1| < 2, \ \operatorname{Im} z < 0 \}$

$w(3) = \infty, \quad w(-1) = 0$

$w(1-2i) = \dfrac{2-2i}{-2-2i} = \dfrac{1-i}{-1-i} = \dfrac{(1-i)^2}{-(1+i)(1-i)} =$

$= \dfrac{-2i}{-2} = i$

$w(1+2i) = \dfrac{2+2i}{-2+2i} = \dfrac{1+i}{-1+i} = -i$

$\Rightarrow w : \{|z-1| = 2\} \longrightarrow \{\operatorname{Re} z = 0\}$

$z = x + 0 \cdot i, \quad -1 \leq x < 3 \Rightarrow$

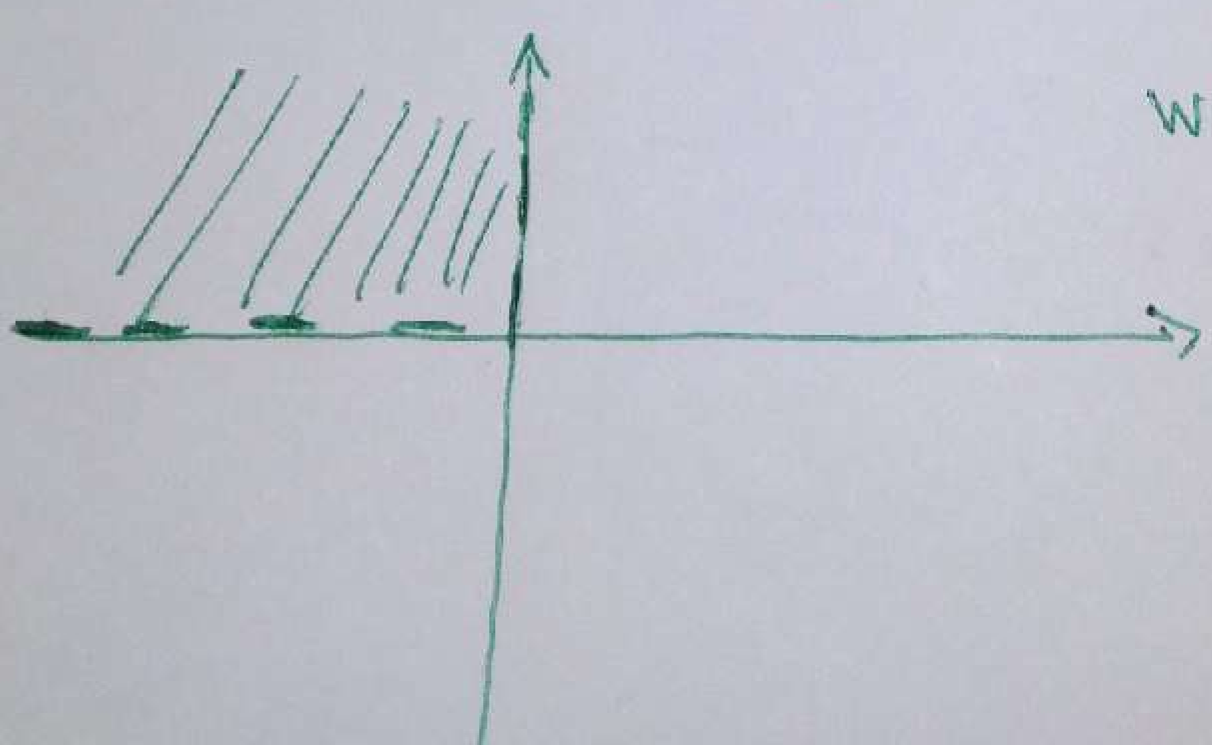
$\Rightarrow -4 < x-3 < 0 \Rightarrow -\infty < \dfrac{1}{x-3} < -\dfrac{1}{4} \Rightarrow$

$\Rightarrow -\infty < \dfrac{4}{x-3} < -1 \Rightarrow -\infty < \dfrac{4}{x-3} + 1 < 0 \Rightarrow$

$-\infty < \dfrac{x+1}{x-3} < 0 \Rightarrow w : (-1, 3) \to (-\infty, 0)$

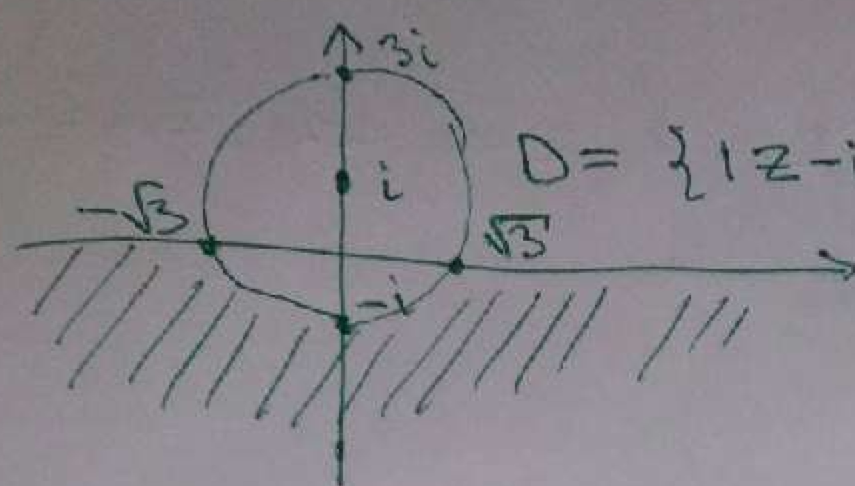
$w(-i) = \dfrac{1-i}{-i-3} = -0{,}2 + 0{,}4i$

$w(D) = \{ \operatorname{Re} z < 0, \operatorname{Im} z > 0 \}$



4.3

$$w = \frac{1}{z}$$



$$D = \{ |z-i| > 2, \operatorname{Im} z < 0 \}$$

$$w(\pm\sqrt{3}) = \pm\frac{1}{\sqrt{3}}$$

$$w : \{(\sqrt{3}, +\infty)\} \to \left(0, \frac{1}{\sqrt{3}}\right)$$

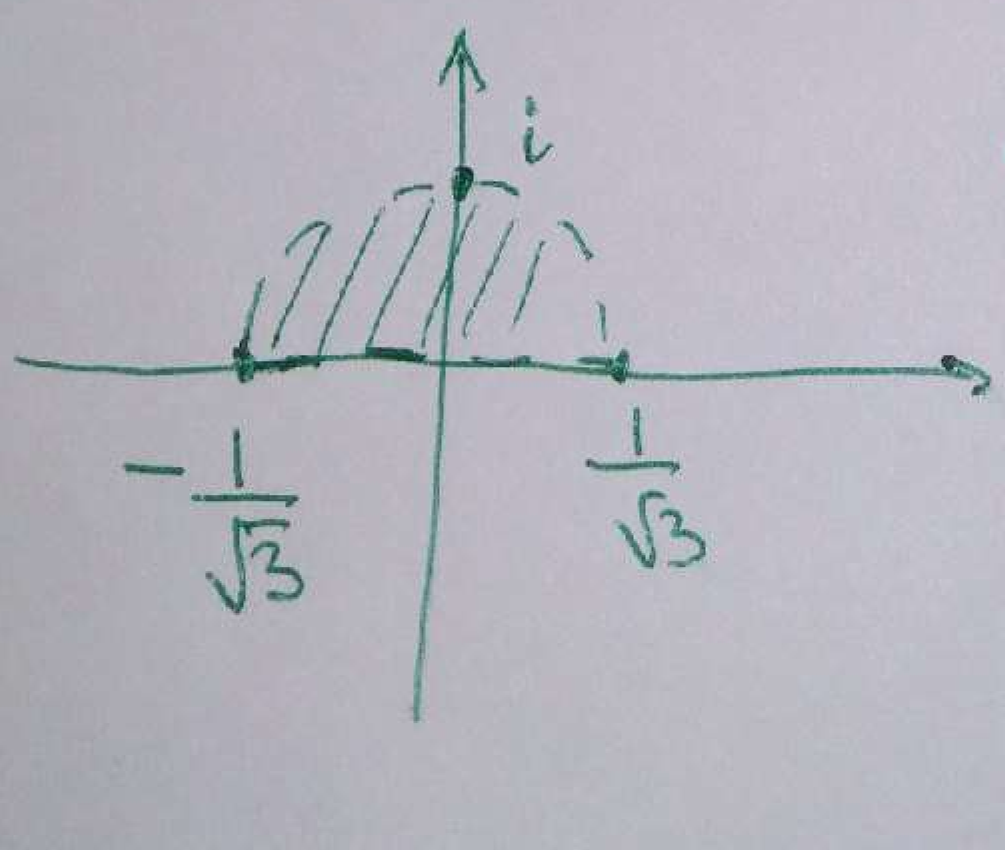
$$w : (-\infty, -\sqrt{3}) \to \cancel{\left(0, \frac{1}{\ }\right.} \left(-\frac{1}{\sqrt{3}}, 0\right)$$

$$\cancel{|z-i| = 2 \Rightarrow z = i}$$

$$w(-i) = \frac{1}{-i} = i$$

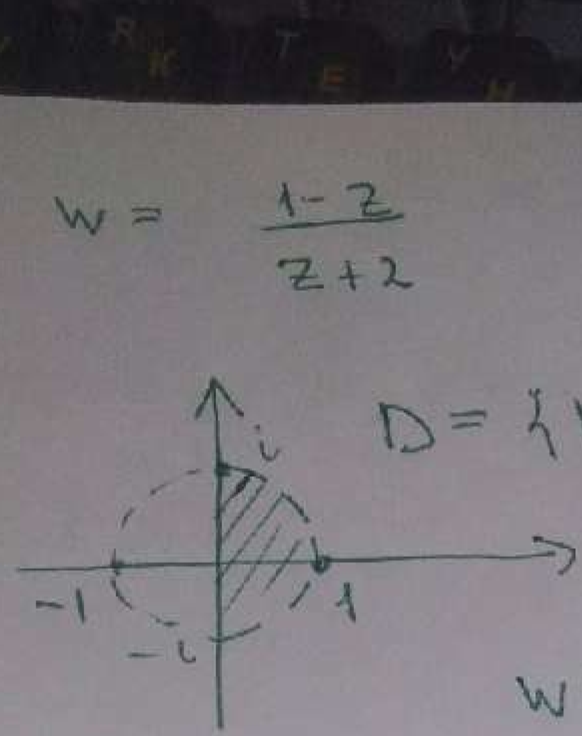
$$w(3i) = \frac{1}{3i} = -\frac{1}{3}i$$

$$w(-2i) = \frac{1}{-2i} = \frac{i}{2}$$



$$w(D) = \left\{ \left|z - \frac{i}{3}\right| < \frac{2}{3}, \operatorname{Im} z > 0 \right\}$$

4.4 $\quad w = \dfrac{1-z}{z+2}$



$D = \{ |z| < 1,\ \operatorname{Re} z > 0 \}$

$w(1) = 0$

$w(0) = \dfrac{1}{2}$

$w(i) = \dfrac{1-i}{i+2} = \dfrac{(1-i)(i-2)}{(i+2)(i-2)} =$

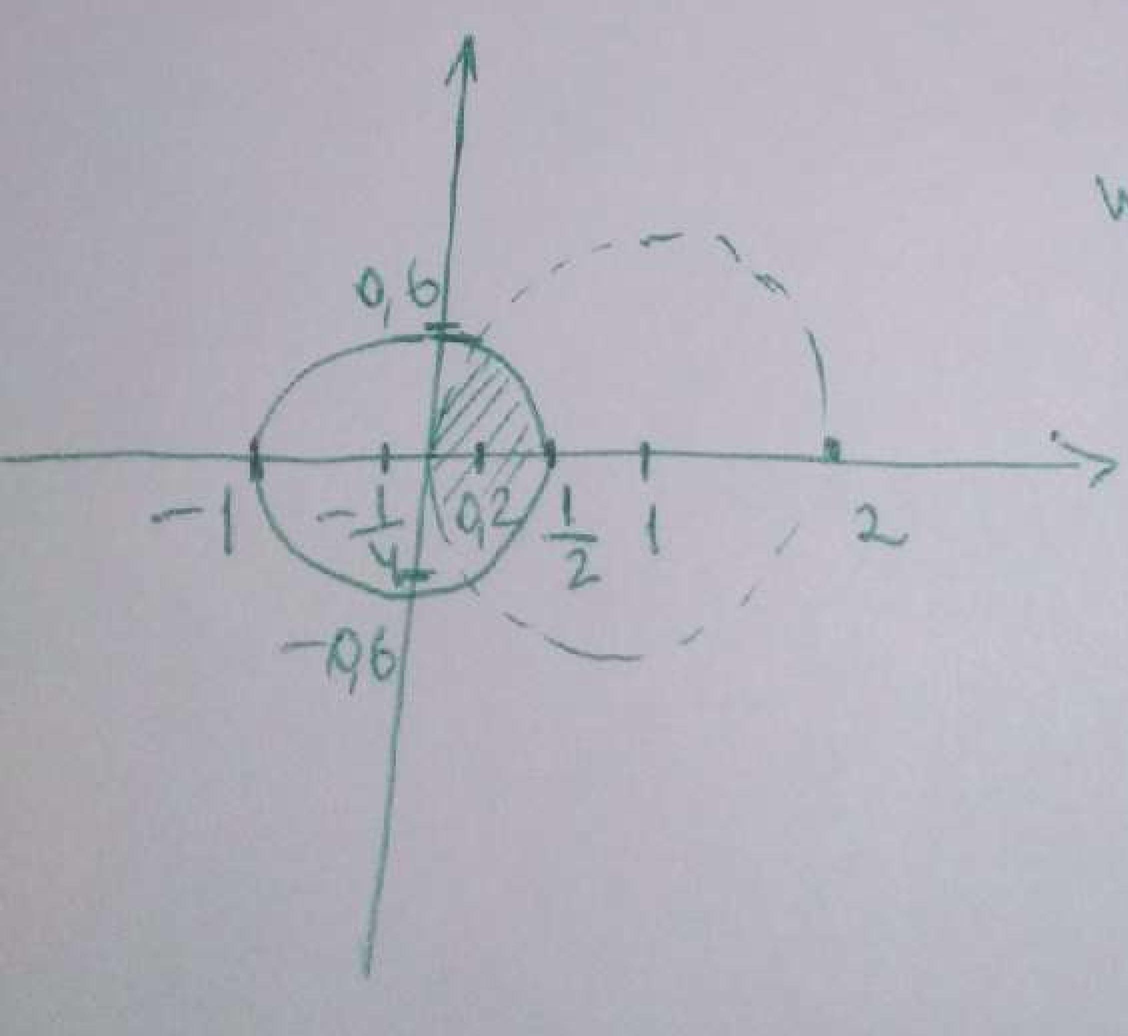
$= \dfrac{1}{5} - \dfrac{3}{5} i$

$w(-i) = \dfrac{1+i}{-i+2} = \dfrac{(1+i)(i+2)}{(-i+2)(i+2)} = \dfrac{1}{5} + \dfrac{3}{5} i$

$w(-1) = \dfrac{2}{1} = 2$

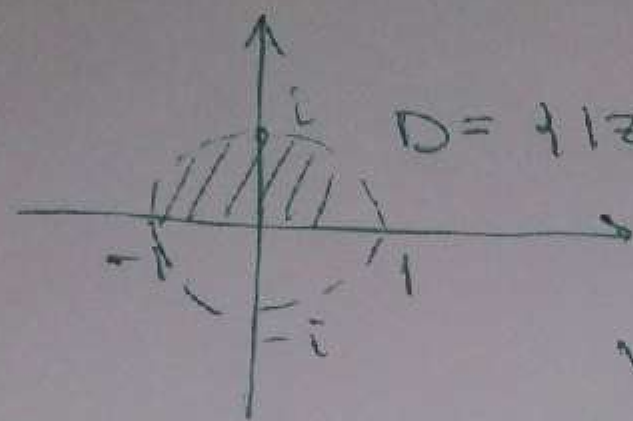
$w(iy) = \dfrac{1-iy}{iy+2} = \dfrac{(1-iy)(2-iy)}{(2+iy)(2-iy)} =$

$= \dfrac{2 - y^2 - 3iy}{4+y^2} = \dfrac{2-y^2}{4+y^2} - \dfrac{3y}{4+y^2} i$



$w(D) = \{ |z-1| < 1,$
$|z + \frac{1}{4}| < \frac{3}{4} \}$

4.5 $\qquad w = \dfrac{1-z}{1+z}$



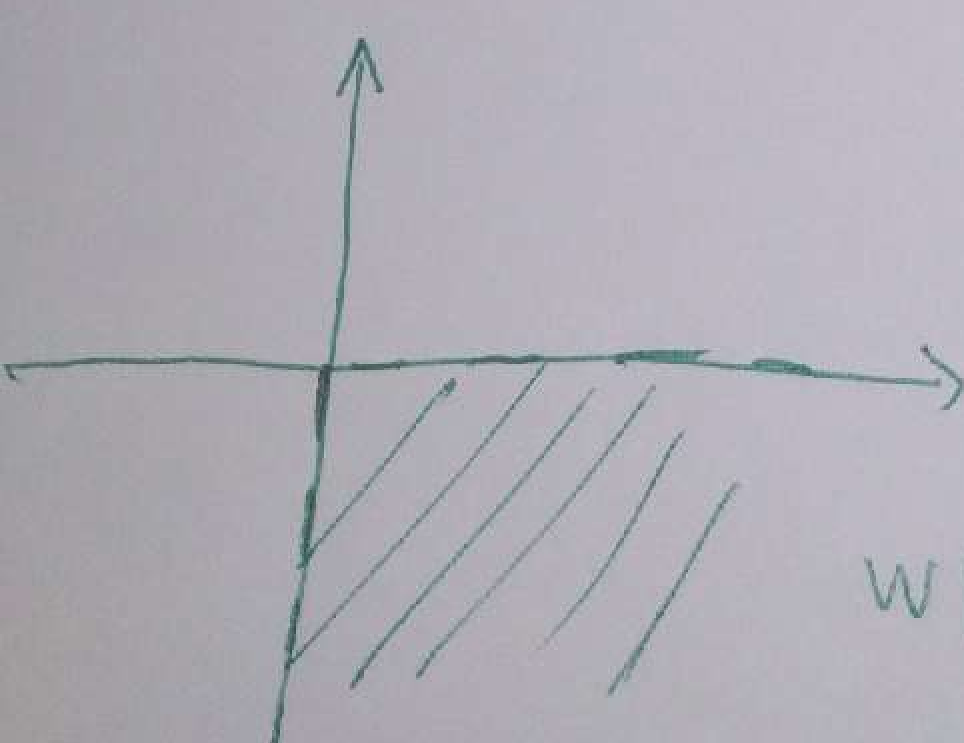
$D = \{|z|<1, \operatorname{Im} z > 0\}$

$$w(-1) = \infty$$
$$w(1) = 0$$
$$w\left(\frac{i}{2}\right) = 0{,}6 - 0{,}8i$$
$$w(i) = \frac{1-i}{1+i} = -i$$
$$w(-i) = i$$

$$w : \{|z| = 1\} \to \{\operatorname{Re} z = 0\}$$

$z = x + 0 \cdot i, \qquad -1 \leq x \leq 1$

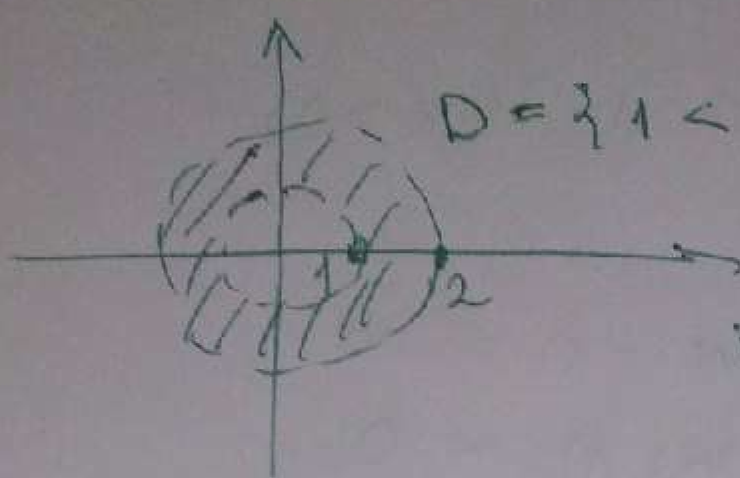
$$0 < 1+x < 2$$
$$\frac{1}{2} < \frac{1}{1+x} < +\infty$$
$$1 < \frac{2}{1+x} < +\infty$$
$$0 < \frac{2}{1+x} - 1 < +\infty$$
$$0 < \frac{1-x}{1+x} < +\infty$$
$$w : (-1, 1) \to (0, +\infty)$$



$$w(D) = \{\operatorname{Re} z > 0, \operatorname{Im} z < 0\}$$

4.6

$$w = \frac{2}{z-1}$$



$D = \{1 < |z| < 2\}$

$$w(1) = \infty$$

$$w(-1) = -1$$

$$w\left(\frac{3}{2}\right) = 4$$

$$w(i) = \frac{2}{i-1} = -1-i$$

$$w(-i) = \frac{2}{-i-1} = -1+i$$

$$\Rightarrow \quad w: |z| = 1 \longrightarrow \operatorname{Re} z = -1$$

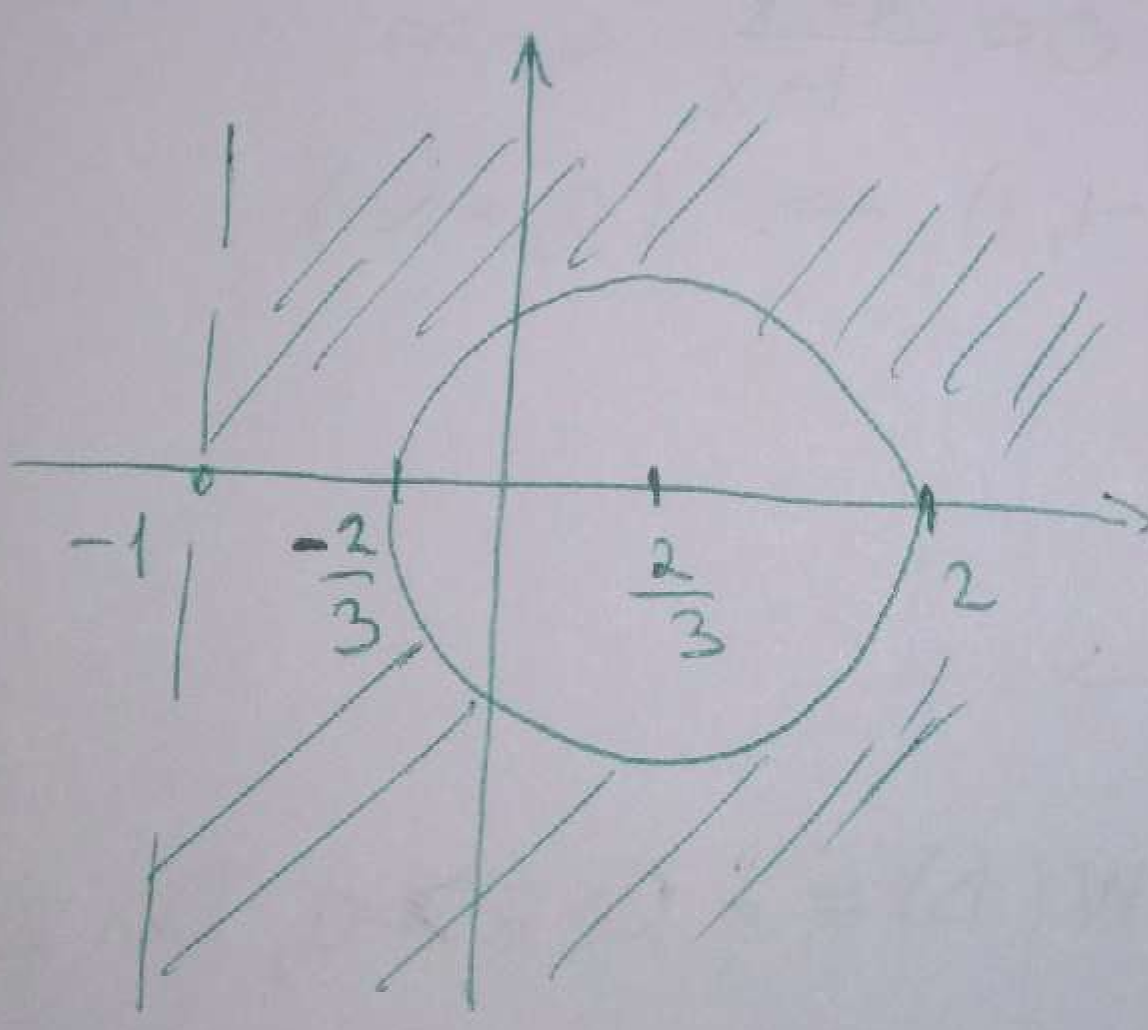
$$w(2) = 2, \quad w(-2) = -\frac{2}{3}$$

$$w(2i) = \frac{2}{2i-1} = -\frac{2}{5} - \frac{4}{5}i$$

$$z_0 = \frac{2 - \frac{2}{3}}{2} = \frac{2}{3}$$

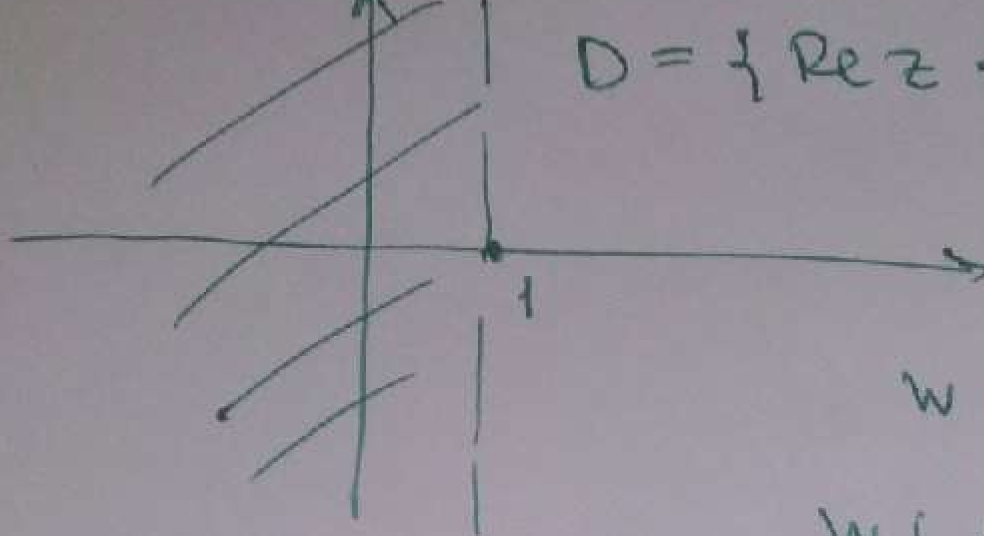
$$r = \frac{2 + \frac{2}{3}}{2} = \frac{4}{3}$$

$$w: |z| = 2 \longrightarrow \left|z - \frac{2}{3}\right| = \frac{4}{3}$$



$$W(D) = \left\{\operatorname{Re} z > -1, \ \left|z - \frac{2}{3}\right| > \frac{4}{3}\right\}$$

4.7 $\quad w = \dfrac{z}{z-2}$



$D = \{\operatorname{Re} z < 1\}$

$w(0) = 0$

$w(1) = -1$

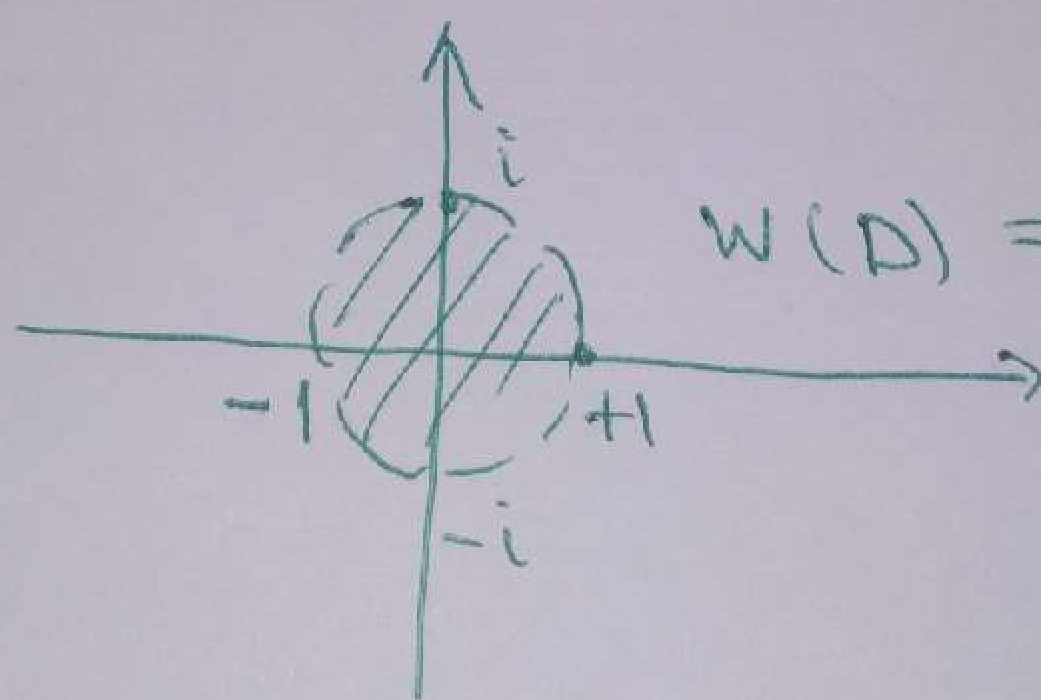
$w(1+i) = \dfrac{1+i}{-1+i} = -i$

$w(1-i) = \dfrac{1-i}{-1-i} = i$

$z_0 = 0$

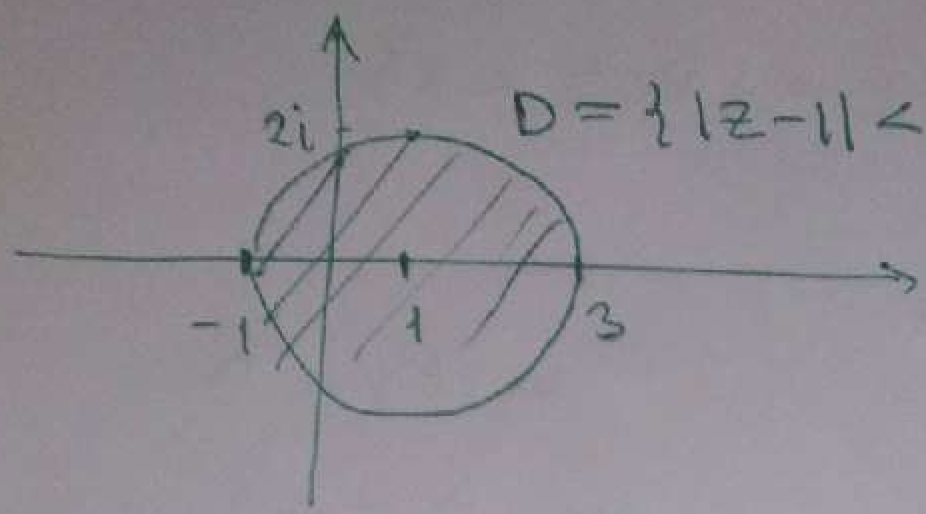
$r = 1$

$w : \{\operatorname{Re} z = 1\} \longrightarrow \{|z| = 1\}$



$w(D) = \{|z| < 1\}$

4.8 $\quad w = \dfrac{z+1}{z-2}$



$w(3) = 4 \quad , \quad w(2) = \infty$

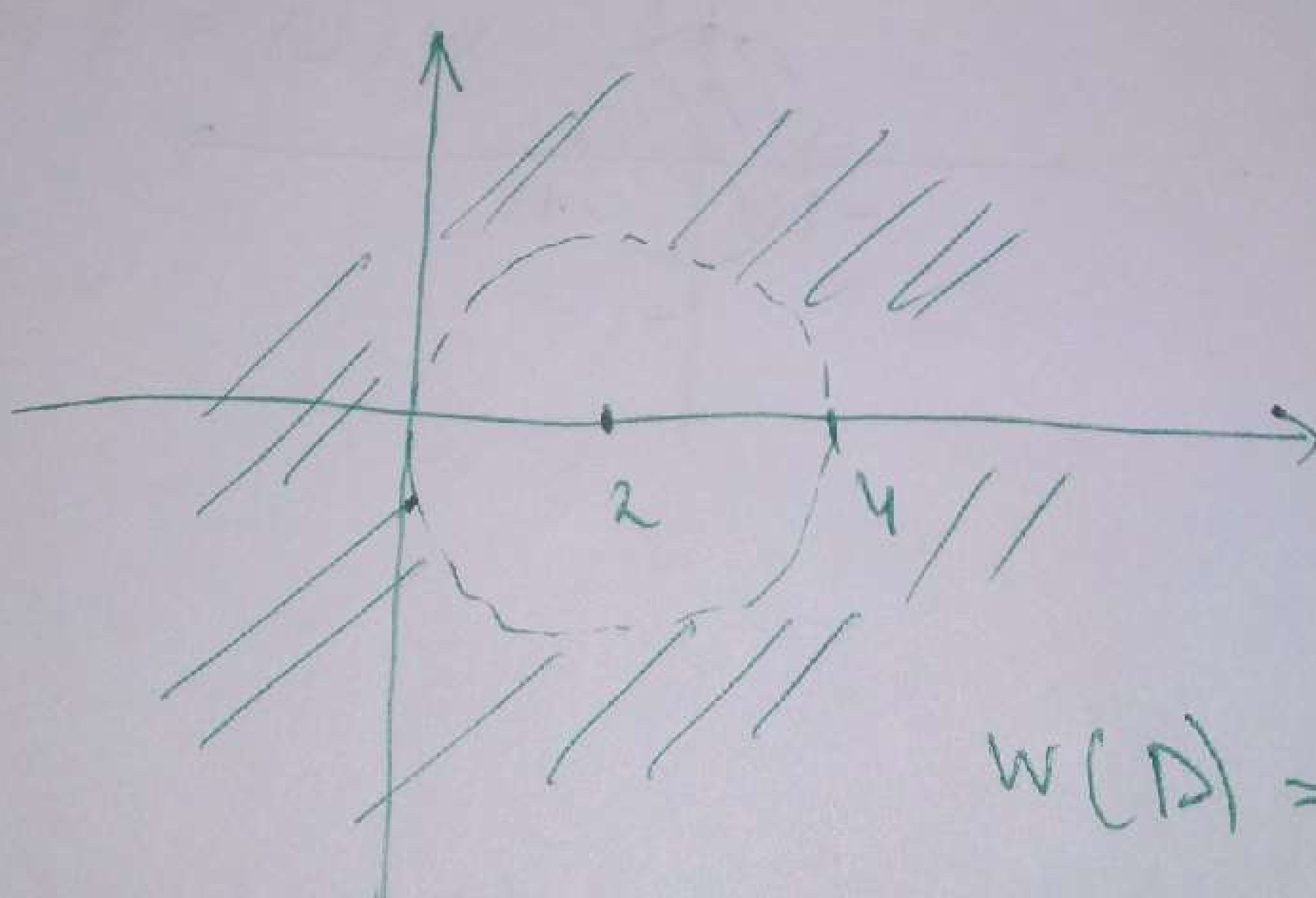
$w(-1) = 0$

$w(1+2i) = \dfrac{2+2i}{-1+2i} = \dfrac{2}{5} - \dfrac{6}{5}i$

$z_0 = \dfrac{4+0}{2} = 2$

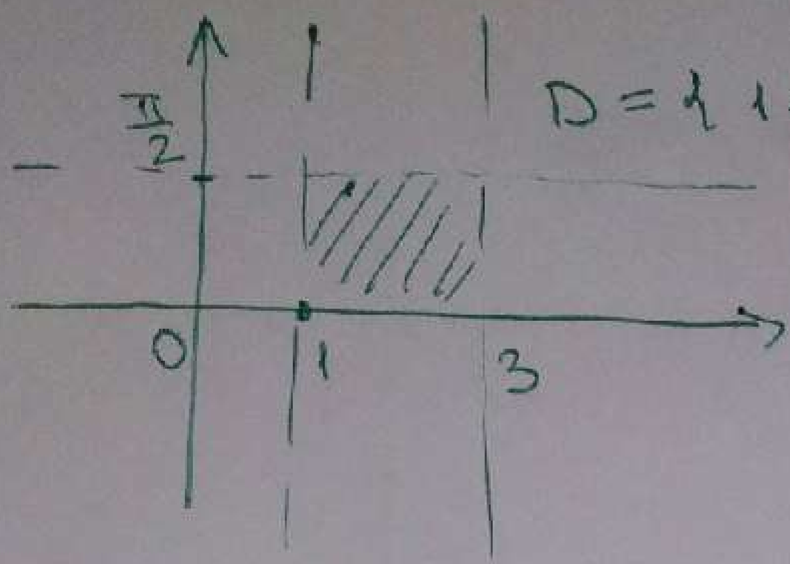
$r = \dfrac{4-0}{2} = 2$

$w : \{|z-1| = 2\} \to \{|z-2| = 2\}$



$w(D) = \{|z-2| > 2\}$

4.9 $w = e^z$



$D = \{1 < \operatorname{Re} z < 3,\ 0 < \operatorname{Im} z < \frac{\pi}{2}\}$

~~$z = x_0 +$~~ $z = 1 + yi,\ 0 < y < \frac{\pi}{2}$

$w(1+yi) = e^{1+yi} = e \cdot e^{yi},\ R = e$

$w(3+yi) = e^{3+yi} = e^3 \cdot e^{yi},\ R = e^3$

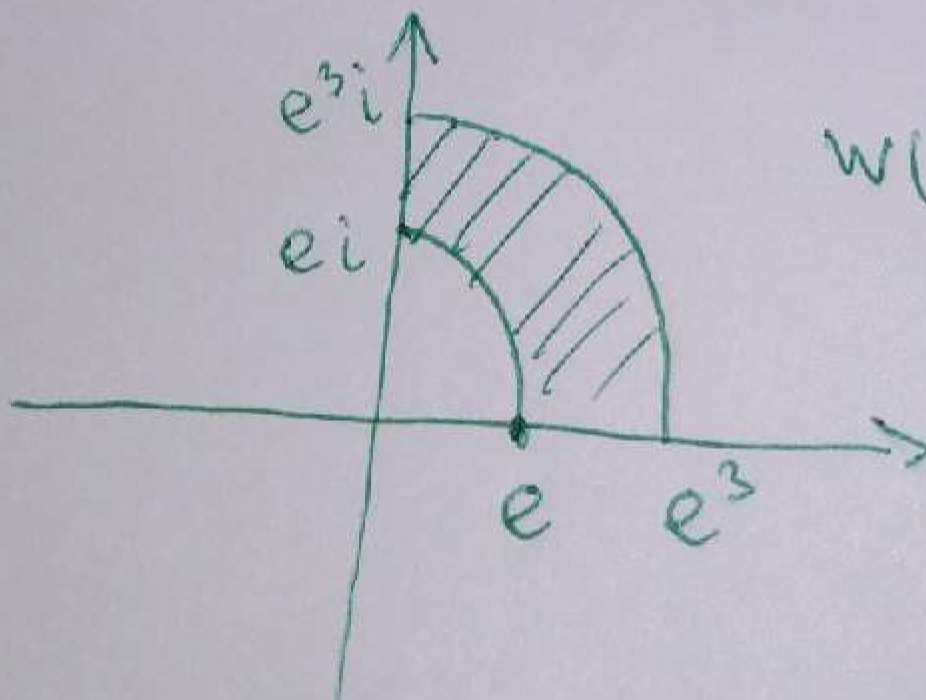
$z = x + 0 \cdot i,\ x \in (1,3)$

$w(x + 0i) = e^{x+0i} = e^x$

$e < e^x < e^3$

$z = x + \frac{\pi}{2} \cdot i,\ w\left(x + \frac{\pi}{2} i\right) = e^{x + \frac{\pi}{2} i} = e^x \cdot i$

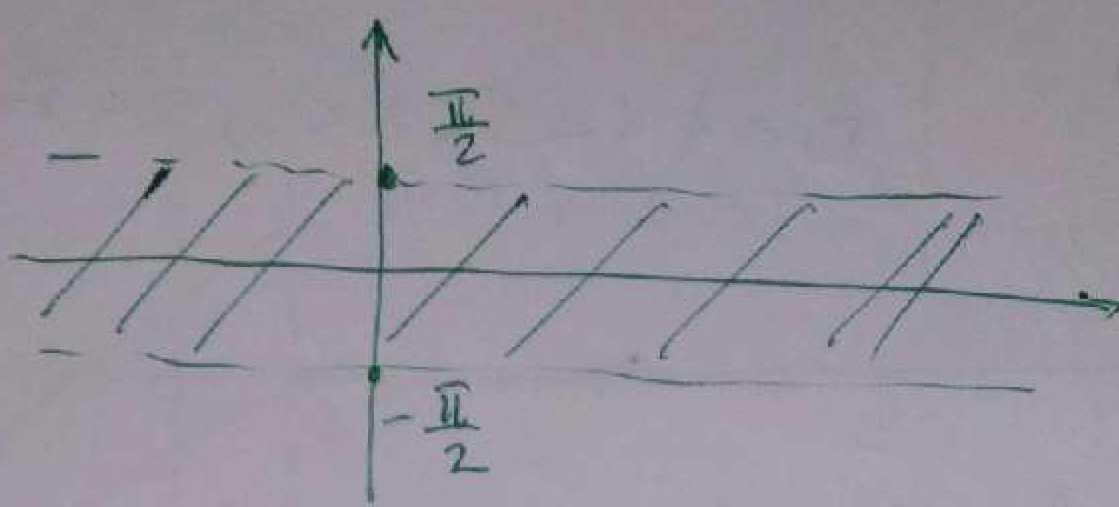
$e \cdot i < e^x \cdot i < e^3 i$



$w(D) = \{e < |z| < e^3,$

$\operatorname{Im} z > 0, \operatorname{Re} z > 0\}$

4.10 $D = \{ |\operatorname{Im} z| < \frac{\pi}{2} \}$, $w = e^z$



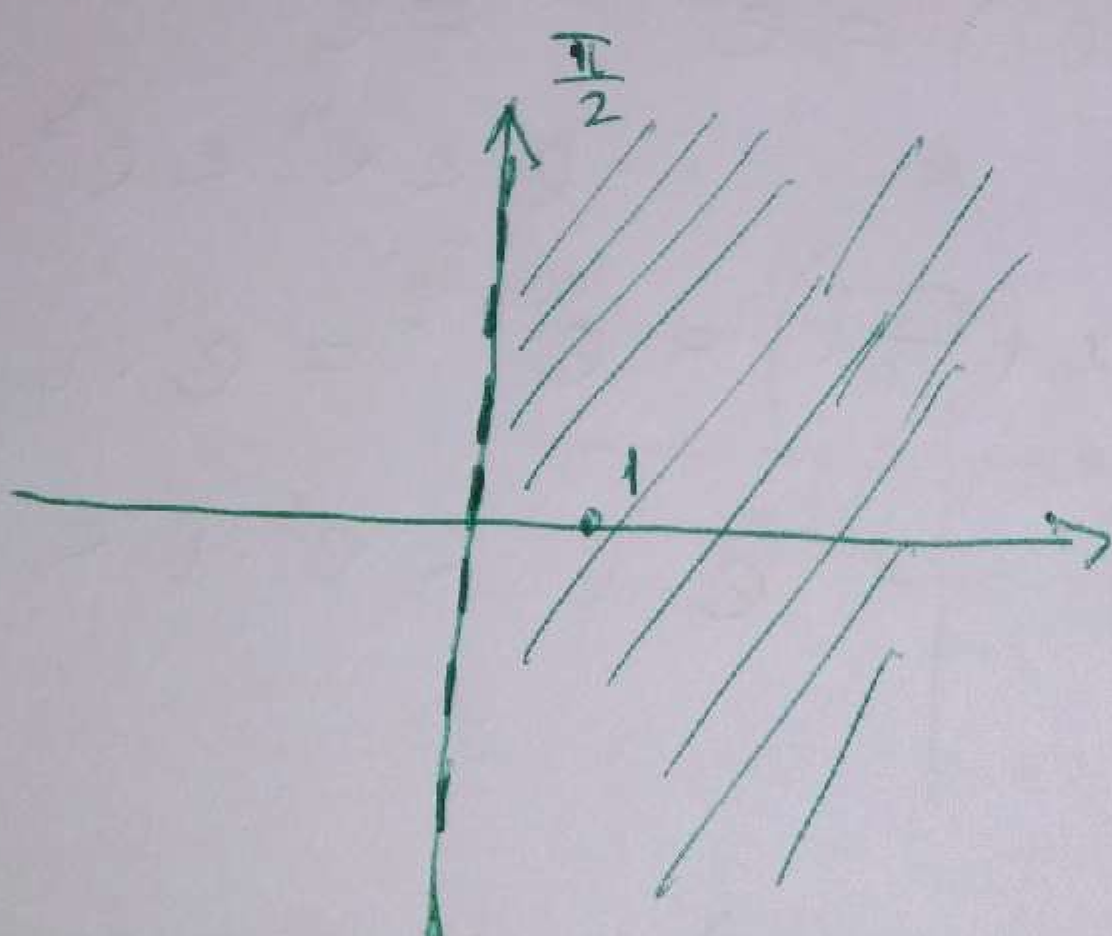
$z = x + \frac{\pi}{2} i$, $x \in (-\infty, +\infty)$

$w(x + \frac{\pi}{2} i) = e^{x + \frac{\pi}{2} i} = e^x \cdot i$, $0 < e^x < +\infty$

$z = x - \frac{\pi}{2} i$, $x \in (-\infty, +\infty)$

$w(x - \frac{\pi}{2} i) = e^{x - \frac{\pi}{2} i} = -e^x \cdot i$, $-\infty < -e^x < 0$

$w(0) = e^0 = 1$



$w(D) = \{ \operatorname{Re} z > 0 \}$